

Lesson 27 — Monster Snack Time (pre-trained classifier)

Matches LEGO Teacher Portal lesson: "Monster Snack Time".

Objective (I can...): I can program actions using a pre-trained classifier's decisions. **CS concept:** Using classifier output in conditionals · **Builds on:** Lesson 26 (classifier events). *Privacy: on-device camera, no logs, images never saved.*

AI Connection (2 min): The classifier's guess feeds an IF/THEN — exactly the "IF I'm sure, THEN act" idea from the conditionals unit, now powered by AI. *First-principles question:* "Should the monster still eat if the AI is only a little bit sure?"

Materials: LEGO CS & AI kit + device per pair with camera, the lesson's food/object cards. **Model reuse:** Monster model from Lesson 26 (small reconfig into Snack Time if the build calls for it) — mostly code.

60-minute flow - Warm-up (8): "Yes food / no food" — pairs sort item cards the monster would or wouldn't eat. - **Concept + demo (10):** Demo the classifier recognizing an item and an if/then choosing the monster's reaction (eat vs. refuse). - **Build/Code (30):** Pairs program conditionals on the model: if the classifier recognizes a "snack," the monster eats (moves + sound); else it reacts differently. *Swap at 15 min.* - **Challenge (7):** Add a score variable that counts snacks eaten. - **Share + pack-up (5):** Pairs demo both reactions and explain the condition. Model stays built.

Pair roles: Driver codes the if/then, Navigator presents items and reads the label. Swap at 15 min. (*Trio: third owns the score add.*)

Differentiation: *Support* — the "if" branch provided; students add the "else." *Extension / fast-finisher* — track two categories with two reactions. *Reteach* — narrate "the AI says snack → true → eat" as they test.

Exit check: Pair's model reacts differently based on what the classifier recognizes. **At-home (optional):** Draw two things the monster would eat and two it wouldn't, and why. ~10 min.
